

Long-distance coherent quantum communications in deployed telecom networks

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Recent advances in quantum communications have underscored the crucial role of optical coherence in developing quantum networks. This resource, which is fundamental to the phase-based architecture of the quantum internet¹, has enabled the only successful demonstrations of multi-node quantum networks^{2–4} and substantially extended the range of quantum key distribution (QKD)⁵. However, the scalability of coherence-based quantum protocols remains uncertain owing to the specialized hardware required, such as ultra-stable optical cavities and cryogenic photon detectors. Here we implement the coherence-based twin-field QKD protocol over a 254-kilometre commercial telecom network spanning between Frankfurt and Kehl, Germany, achieving encryption key distribution at 110 bits per second. Our results are enabled by a scalable approach to optical coherence distribution, supported by a practical system architecture and non-cryogenic single-photon detection aided by off-band phase stabilization. Our results demonstrate repeater-like quantum communication in an operational network setting, doubling the distance for practical real-world QKD implementations without cryogenic cooling. In addition, to our knowledge, we realized one of the largest QKD networks featuring measurement-device-independent properties⁶. Our research aligns the requirements of coherence-based quantum communication with the capabilities of existing telecommunication infrastructure, which is likely to be useful to the future of high-performance quantum networks, including the implementation of advanced quantum communication protocols, quantum repeaters, quantum sensing networks and distributed quantum computing⁷.

Quantum communications exploit the properties of quantum-mechanical states to enable applications that are provably impossible using classical means alone. These applications include information-theoretic secure communication^{8–11}, exponential speed-up in communication^{12–14}, quantum sensor networks^{15–18}, networked quantum computing^{1,19–21} and more^{22–25}. Quantum key distribution (QKD)^{8,10,11} is the most developed of these, providing information-theoretic secure key distribution over untrusted public channels without algorithmic assumptions and regardless of the resources and capabilities of potential attackers. Owing to their long-distance transmission capability, photons are the preferred carriers for quantum communications and are typically encoded as qubits, the elemental unit of quantum information. When quantum links connect multiple quantum-capable nodes, a quantum network is formed, or, on a larger scale, a ‘quantum internet’.

Coherent quantum communications, a subset of quantum communications, requires maintaining optical coherence between nodes^{26,27}. Optical

coherence is a powerful resource, supporting various multi-node quantum applications^{14,15}. Notably, it underpins the Duan–Lukin–Cirac–Zoller teleportation mechanism²⁸, which is one of the most practical implementations of a fundamental primitive for the quantum internet²⁹. This mechanism has enabled the first three-node quantum networks^{2,4} and quantum teleportation across non-adjacent nodes of a network³.

When used in QKD, optical coherence facilitates protocols such as twin-field QKD (TF-QKD)^{5,30,31} and post-measurement pairing QKD^{32,33}. These are the most practical ways to overcome the repeaterless secret key capacity bound (SKC₀)³⁴, the fundamental capacity of a quantum channel not endowed by a quantum repeater¹¹. They also double the maximum communication distance compared with standard point-to-point QKD protocols, expanding the reach of fibre-based QKD to national and international distances. Several TF-QKD lab demonstrations³⁵ have achieved distances exceeding hundreds of kilometres^{36–40}, with one reaching up to 1,000 km (ref. 41).

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Table 1 | Comparison of typical properties of APDs and SNSPDs

	APDs	SNSPDs
Detection efficiency (%)	10–20	60–80
Dark counts	Hundreds of hertz	A few hertz
Afterpulse (%)	About 3	0
Size (U)	<1	>10
Cost (US\$)	1,000s	100,000s
Cooling system	Thermoelectric (20 to –60) °C	Cryogenic (–270 to –272) °C

The cost indicates the order of magnitude for an entire APD or SNSPD system. The system sizes are expressed in rack units (U), the typical unit of measure for telecom rack-mounted equipment: 1U corresponds to 44.45 mm thickness.

Quantum communications depends on appropriate channels and nodes to transmit and process quantum states. The widespread adoption of the quantum internet is contingent on aligning quantum communications implementation requirements and practices with the capabilities of existing telecommunication infrastructures. At present, integration has been achieved for only simpler, point-to-point, non-coherent QKD systems. Coherence-based protocols have been primarily tested in laboratories with specialized equipment such as ultra-stable optical cavities and cryogenic photon detectors^{37–43}, which are unsuitable for typical telecommunication environments. Given the benefits of using optical coherence in this domain, developing methods to integrate it with existing telecommunication frameworks is crucial for the advancement of the field and for large-scale adoption.

This study demonstrates the integration of coherent quantum protocols into commercial telecommunication infrastructure, linking colocation data centres in Germany via optical fibres with high losses and asymmetric link lengths. The system implemented the TF-QKD protocol⁵ and incorporated practical technologies that facilitate seamless operation.

Implementing coherent, or phase-based, quantum communications poses several challenges such as establishing a common phase-reference frame among distant encoding users^{26–28} and mitigating the phase noise stemming from lasers and transmission channels. To address these challenges, we developed a practical architecture for optical frequency dissemination and orchestration among network nodes and active off-band phase stabilization using non-cryogenically cooled detectors.

In our system, the central node distributes two optical frequency references to the transmitting nodes through a service fibre, allowing them to lock their lasers to a common frequency reference and achieve mutual phase-locking. This eliminates laser phase noise more practically and cost-effectively than using ultra-stable lasers and external cavities^{37,40–42}. To mitigate phase noise from the fibre, we use an off-band phase-stabilization feedback system³⁶ based on single-photon interference outcomes monitored by avalanche photodiodes (APDs). This combination of technologies is crucial for our experiment’s success.

Although APDs offer practical semiconductor-based single-photon detection capabilities, they have less ideal properties compared with superconductive nanowire single-photon detectors (SNSPDs), the detection technology used to support phase-based quantum communications so far^{2–4,37,39,41}. As summarized in Table 1, APDs show higher dark counts, lower detection efficiency and susceptibility to afterpulse effects. However, APDs are one to two orders of magnitudes more economical than SNSPDs, are more practical and can operate at temperatures compatible with telecommunication infrastructure.

The success of this implementation hinges on combining the off-band stabilization technique with APDs. For long-distance coherent quantum links, distributing significantly brighter phase-reference signals than the protocol-encoded ones is crucial to establishing a

common phase-reference frame among users. Previous methods, which used time-multiplexed phase-reference pulses at the same optical frequency as the encoded signals, are incompatible with APDs owing to the afterpulse effect^{44–46}. Using the same APDs to detect intense reference pulses and protocol-encoded signals introduces noise that obscures the encoded signals. By contrast, our off-band stabilization employs distinct optical frequencies for phase-reference and protocol-encoded signals, allowing separate detectors for each, eliminating afterpulse cross-contamination.

System details and methods

The experiment was conducted in Germany, with access to the network infrastructure provided by GÉANT. Communication was established over a link spanning 253.9 km between Frankfurt and Kehl with a 56.0-dB loss and a relay in Kirchfeld, approximately three-fifths of the way. This set-up forms a star-shaped quantum network with three nodes: two transmitters, Alice and Bob, at the network’s edge, and a central relay receiver, Charlie. Charlie is connected to each transmitter by a fibre duplex cable (characterization is provided in the Supplementary Information). Equipment was housed within standard telecom racks in colocation data centres, functioning alongside existing telecommunication gear. The network’s geographical and functional layout, including the fibre links and system elements, is detailed in Fig. 1.

The transmitters prepared optical weak coherent pulse (WCP) qubits for the quantum communication protocol and the receiver node distributed common optical frequency references to ensure coherence and interfered the WCPs from transmitters. Each node features a modular design with interconnected 19-inch rack boxes, enhancing compatibility with telecom racks and system flexibility. The nodes’ subsystems are organized into three functional layers: service, management and quantum, each spanning across the three nodes, as shown in the top diagram of Fig. 1.

The service layer facilitates remote operation and intra-node communication, and distributes two optical frequency references (λ_s and λ_Q) from Charlie to the transmitters along with a clock for synchronization of electronics. It comprises two master lasers (L_s and L_Q), service fibres (F_C^{AC} and F_C^{BC}), ethernet switches equipped for optical networking, a dense wavelength division multiplexing (DWDM)-based optical routing set-up and other supporting components.

The management layer controls all system equipment. It includes a server (Server^{A,B,C}) and a custom-programmed field-programmable gate array (FPGA^{A,B,C}) per node, and an arbitrary waveform generator (AWG) for each transmitter.

A description of the components and functionalities implemented by the service and management layers is reported in the Methods, and a simplified representation of the routing of the optical signals in the system is shown in Fig. 2. For a comprehensive schematic of the system, refer to the Supplementary Information.

The quantum layer executes the quantum communication protocol. The specific protocol implemented in this experiment is the sending–not-sending variant^{30,47} of TF-QKD⁵. This protocol enables continuous generation of a shared secret bit string between Alice and Bob, enabling quantum-safe communication by means of symmetric-key cryptography. The transmitters encode information on the optical phase of WCPs, which are sent to Charlie, where they interfere. Charlie monitors the interference outcome with two single-photon detectors and announces the results (which detector clicked and when) on a public channel. Charlie’s measurements constitute a Bell-state measurement of the joint quantum system composed of Alice’s and Bob’s WCP qubits^{6,26,27}. By combining the publicly announced interference results with their private knowledge of the WCPs encoding, the transmitters initiate a sifting procedure (described in the Methods) that generates a shared secret bit string. The TF-QKD protocol ensures measurement-device independence for the detections made by Charlie^{6,48}, and a favourable

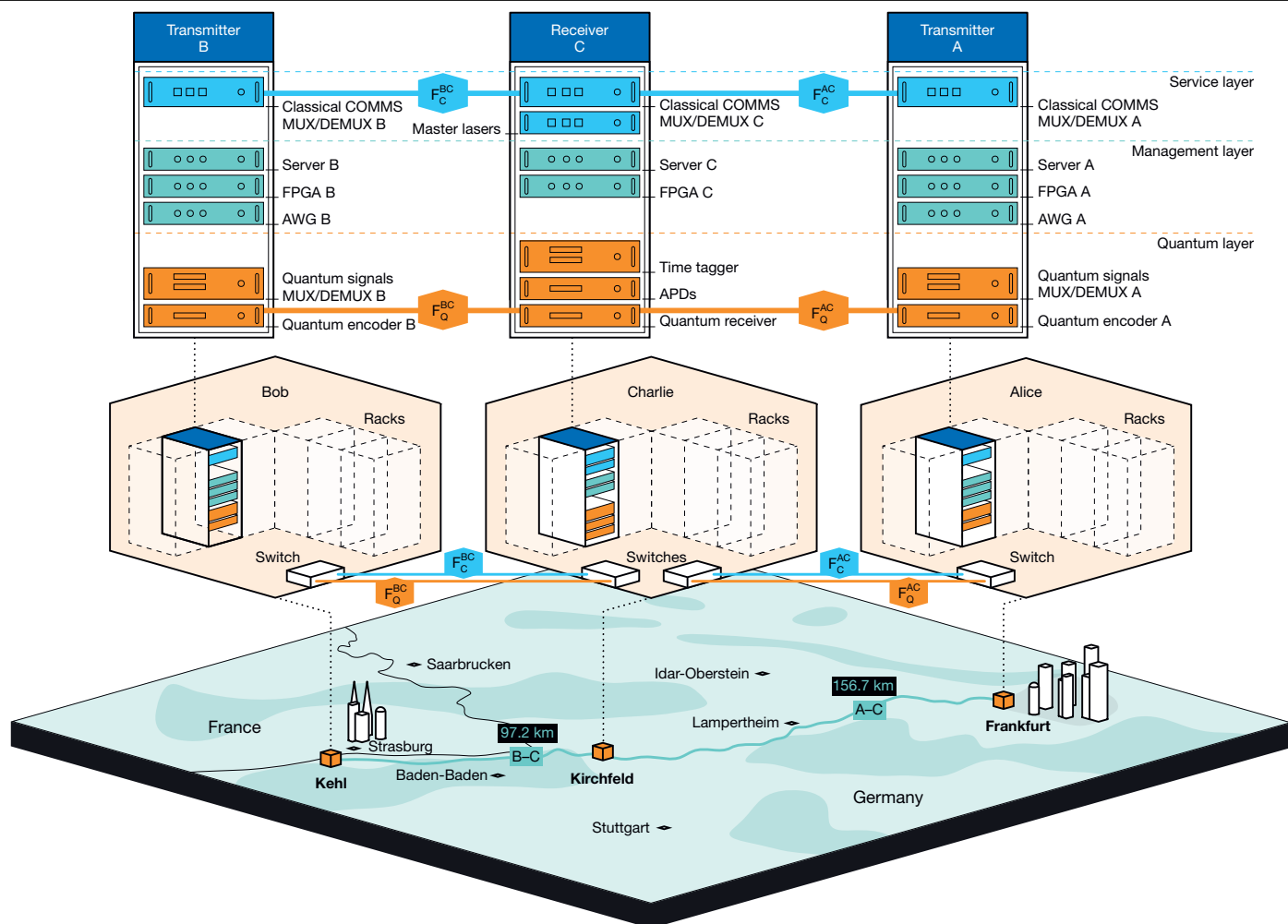


Fig. 1 | Deployed coherent quantum communications system. The TF-QKD system consists of three nodes: two transmitter nodes, Alice and Bob, and one receiver node, Charlie (called A, B and C, respectively). Bottom: equipment for nodes A, B and C was installed in three colocation data centres located in Frankfurt, Kehl and Kirchfeld alongside other running telecommunication equipment. All equipment was mounted in 19-inch telecom racks, and the 3 nodes were interconnected pairwise through 2 fibre duplexes. The A–C fibre

duplex spans 156.7 km, with the 2 bundled fibres, F_Q^{AC} and F_C^{AC} , characterized by losses of 32.4 dB and 33.5 dB, respectively. The B–C duplex spans 97.2 km, with F_Q^{BC} and F_C^{BC} characterized by losses of 21.7 dB and 22.2 dB, respectively. Top: functional schematic of the subsystems installed in the three colocation centres, with equipment grouped into three layers spanning the three nodes based on their implemented functionality. COMMS, communications.

square-root scaling of the secret key rate (SKR) with channel loss⁵. The transmitter nodes have two subsystems in the quantum layer, the quantum multiplexer/demultiplexer (MUX/DEMUX) unit and the quantum encoder.

The quantum MUX/DEMUX unit receives two optical frequency references distributed by Charlie (λ_S and λ_Q) from the service layer, separates them, and routes them appropriately for local processing and encoding before launching them into the quantum channel as λ'_S and λ'_Q .

The quantum encoder employs optical injection locking (OIL)^{35,49,50} to frequency-lock a locally controlled laser diode to the frequency reference λ_Q distributed by Charlie. A detailed discussion on the security implications of this method can be found in the Supplementary Information. The laser diode's output is modulated by several intensity modulators and phase modulators arranged in series, which encode 120-ps-long optical pulses at 1 GHz with arbitrary phases in the $[0, 2\pi]$ interval, and 4 possible optical intensities (required for the decoy-state method^{51,52}). The encoded WCPs (also referred to as λ'_Q) are then adjusted for photon flux and polarization and passed back to the quantum MUX/DEMUX unit for transmission to the receiver. Only for Bob, the λ'_Q passes through a fibre stretcher before entering the quantum

MUX/DEMUX unit. The fibre stretcher serves as an actuator for a fine phase-stabilization mechanism, which is described below. Detailed operations of the quantum MUX/DEMUX and encoder units are outlined in the Supplementary Information.

Multiplexing the manipulated λ'_S and λ'_Q into the quantum channels is crucial for the off-band phase-stabilization technique³⁶, which ensures complete phase stability of an optical channel in two steps: coarse stabilization via a phase modulator at the receiver affecting both λ'_S and λ'_Q , and fine stabilization using a fibre stretcher in Bob, specifically targeting λ'_Q . Together, these stabilizations maintain coherence between optical signals prepared at distant nodes. This process is detailed further in the Methods.

The λ'_S and λ'_Q signals from Alice and Bob reach the receiver unit in Charlie via the quantum channels F_Q^{AC} and F_C^{BC} . At the receiver, these signals are processed by Charlie's quantum layer consisting of a quantum receiver, four APDs and a time tagger. Bob's signals undergo coarse phase stabilization via a lithium niobate (LiNbO₃) phase modulator, and then λ'^A and λ'^B signals interfere on a 50/50 beam splitter. The interference outcomes, separated by 2 DWDMs, are monitored by three -30°C -cooled 1-GHz self-differenced APDs⁵³ (Fig. 2). APD₁ and APD₂ monitor the outputs of the λ'_Q signals interference, and their clicks are

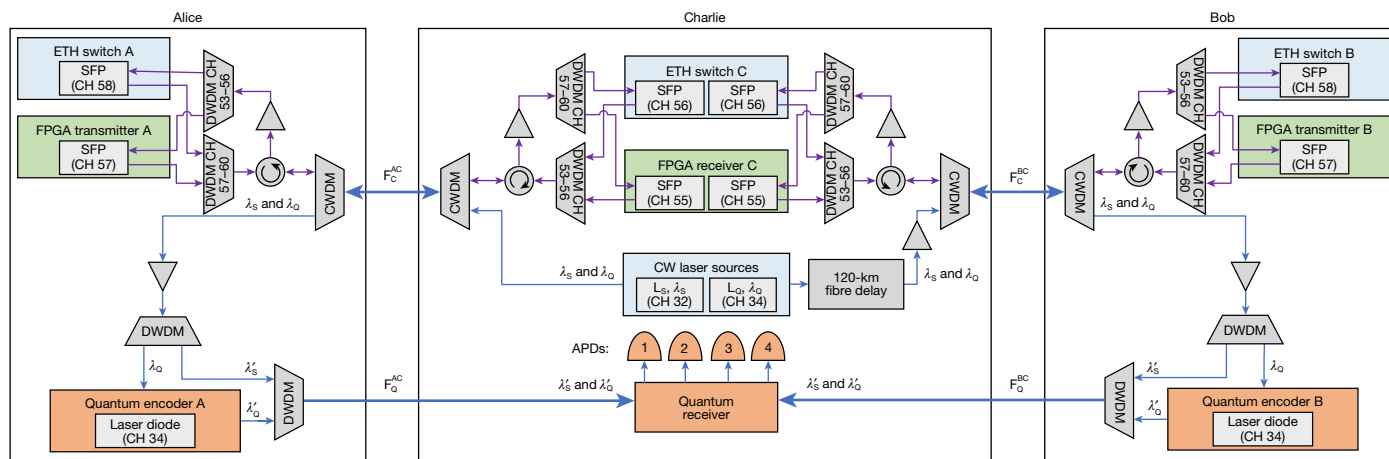


Fig. 2 | Routing of the optical signals. The service fibres F_c^{AC} and F_c^{BC} distribute the optical frequency references λ_s and λ_Q from the central to the transmitter nodes, while also supporting the bidirectional classical communication needed to establish a time-synchronized local network. F_Q^{AC} and F_Q^{BC} transmit the encoded λ'_s and λ'_Q signals from the transmitters to the central node, where

they interfere. A comprehensive schematic of the system is reported in the Supplementary Information. Triangles, EDFAs; circles, circulators; trapezoids, coarse wavelength-division multiplexers (CWDMs) and dense wavelength-division multiplexers (DWDMs). CH, channel; CW, continuous waveform; ETH, ethernet; SFP, small form-factor pluggable transceiver.

collected by a time tagger that records the time of arrival of each photon detection, which is required for the protocol execution. APD_3 monitors the λ'_s signal output, with clicks recorded by this detector utilized to aid the coarse feedback of the off-band phase stabilization.

The protocol execution relies on several automatic stabilization mechanisms within the quantum layer, which are described in detail in the Supplementary Information. Notably, the quantum communication system is designed for remote control and autonomous operation in physically restricted locations to comply with policies of modern deployed telecom infrastructure.

Results

Before the field experiment, we conducted laboratory tests using a test-bed that mimicked the actual field conditions with single-mode-fibre spools and fixed optical attenuators to emulate the distance and channel losses of the installed fibres. These tests provided data acquired in a controlled environment for comparison with field results.

After deploying the system in Germany, we monitored the polarization drift of the installed optical fibres by assessing the optical intensity of the orthogonal polarization component recorded by APD_4 . Field data showed higher polarization stability compared with the lab environment, with minimal intensity drift of the orthogonal polarization component over 12 h, attributed to the natural temperature stability of underground fibres. By contrast, lab-installed fibres, despite being in a temperature-controlled environment, showed more fluctuation owing to less consistent temperature conditions. This is observed for all four signals entering the quantum receiver and is illustrated in Fig. 3 for λ'_s . Our tests also validated our polarization stabilization mechanism, which consistently maintained a 21-dB intensity contrast between the preferred and its orthogonal polarization axes.

In phase-based quantum communication systems¹, maintaining coherence among quantum states encoded by different users is crucial for system performance and error minimization. Optical interference visibility is a reliable metric for benchmarking the consistency and quality of coherence distribution³⁵. In our case, we monitor the interference between Alice's and Bob's λ'_Q signals, as detected by Charlie.

To ensure system coherence, we first evaluated the intrinsic coherence of our laser sources, L_Q and L_s (Fig. 2). Owing to significant link length discrepancies between Charlie to Alice and Bob, we introduced a 120-km-long delay fibre in Charlie's λ_s and λ_Q signals path to Bob,

effectively reducing the optical path mismatch to about 1 km—well within the coherence length of L_Q and L_s , which have a linewidth of a few hundred hertz. More details on this aspect are reported in the Methods.

In addition, we addressed signal degradation caused by the passage through optical components, amplification noise from erbium-doped fibre amplifiers (EDFAs), suboptimal OIL and cross-talk from classical signals in the service layer. These issues were mitigated by continuous OIL optimization at the transmitters, careful EDFA selection, and narrow spectral filtering of λ_s and λ_Q at the transmitters, ensuring maximal interference visibility at the receiver. As a result, the interference visibility between λ_Q^A and λ_Q^B free-drifting signals is 99.09%, indicating robust coherence maintenance of our complex set-up.

Furthermore, when coarse stabilization is active, interference visibility drops slightly owing to noise from λ'_s signals and phase noise from coarse feedback, measuring at 97.15% in the field. This visibility supports effective coherent quantum communication with a low error rate.

In phase-based quantum communications, it is crucial to counterbalance the phase noise introduced by optical fibres for successful protocol execution³⁶. The standard deviation of the phase drift, characterizing this noise, is measured in the field as $\sigma_{UN}^{field} = 8,256 \text{ rad s}^{-1}$ and $\sigma_{UN}^{lab} = 9,197 \text{ rad s}^{-1}$ in the lab. These measurements indicate similar phase behaviours in both environments, with further details provided in the Supplementary Information.

The phase noise from the fibres is mitigated using an off-band phase-stabilization mechanism. Figure 4 shows how the interference between λ'_Q signals evolves through different stabilization stages. Initially, with no stabilization activated, fast intensity fluctuations occur. Upon activating coarse phase stabilization, the phase drift is significantly reduced to $\sigma_{PS} = 2.2 \text{ rad s}^{-1}$. Full stabilization is achieved with fine stabilization activated over λ'_Q , reducing the phase drift to $\sigma_{PS} = 0.47 \text{ rad s}^{-1}$ and locking the phase difference between the two arms to a target value with an uncertainty below 0.1 rad.

Minimizing the background noise detected by APD_1 and APD_2 is essential for successful protocol execution in our quantum communication system. Key noise sources include APD dark counts, afterpulse effects and suboptimal vacuum-state encoding. The dark-count rate and afterpulse effect are inherent properties of the detectors that depend on their structure and operation conditions. Differently from the dark counts, the detrimental impact of the afterpulse on the key generation

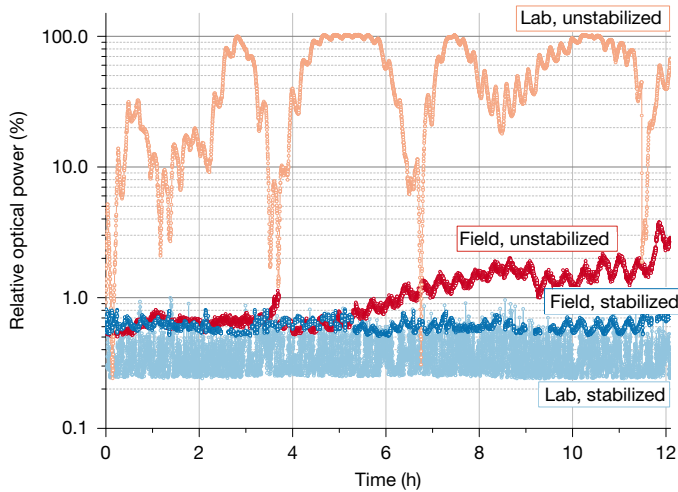


Fig. 3 | Polarization drift and stabilization. Comparison of optical intensities (relative to their maximum values) associated with polarization rotations introduced by the channels. The graph illustrates the orthogonal components of $\lambda_{\text{S}}^{\text{A}}$ recorded in the lab or the field by APD₄. These signals travel over a 313.4-km-long channel, equivalent to twice the A–C distance owing to the looping-back of the support signal by the transmitters. The effect of polarization rotation drift measured in the lab (pink) greatly exceeds that measured in the field (red). Spontaneous polarization rotations introduced by the installed fibres resulted in an intensity drift of the orthogonal polarization component from its minimum value to only 4% of its maximum value over 12 h. In the lab testbed, the same signal fluctuated four times between its minimum and maximum values over the same period. Upon activation of the polarization stabilization mechanism, it effectively maintains the intensity of the orthogonal polarization below 0.8% of its maximum value both in the lab (cyan) and in the field (blue).

can be mitigated by setting a long deadtime window to disregard subsequent counts following an initial successful detection^{46,54}. In this experiment, we set a deadtime window of 5 μs , eliminating the impact of afterpulsing on key generation.

The system implemented the actively odd-parity pairing sending-or-not-sending (SNS) TF-QKD protocol⁴⁷ at a 1-GHz repetition rate. Pulses were interleaved with a 50% duty cycle, alternating between key generation WCP and phase-unmodulated pulses for fine phase stabilization. Protocol-encoded pulses were prepared in two bases. The $\mathbb{Z}$ (coding) basis includes phase-randomized pulses of two intensities, s and n , representing the sending and not-sending bits of the protocol. The $\mathbb{X}$ (testing) basis includes phase-encoded pulses of three levels, u , v and w , and was used to implement the decoy-state method⁵¹. Pulse intensities were independently set for s , u and v , whereas n and w were matched, yielding four intensity settings per transmitter. Pulse intensities and distribution were optimized to maximize key generation in asymmetric link conditions, and to meet crucial security conditions for asymmetric protocol use⁵⁵.

The system achieved an SKR of 2.20×10^{-7} bits per pulse, equivalent to 110.1 bit s^{-1} over 7.5 h of uninterrupted operation. Over this period, the quantum bit error rate for the $\mathbb{X}$ basis averaged 4.61% for u pulses and 5.84% for v pulses. For a comprehensive discussion of the patterns and results, refer to the Supplementary Information.

Discussion

The obtained SKR is depicted in Fig. 5, alongside fundamental bounds for point-to-point QKD. The SKR surpasses the upper bound for the noisy SKC₀ (yellow line in Fig. 5), which can be interpreted as the upper bound for point-to-point QKD communications employing detectors with the same efficiency and dark counts as those used in this

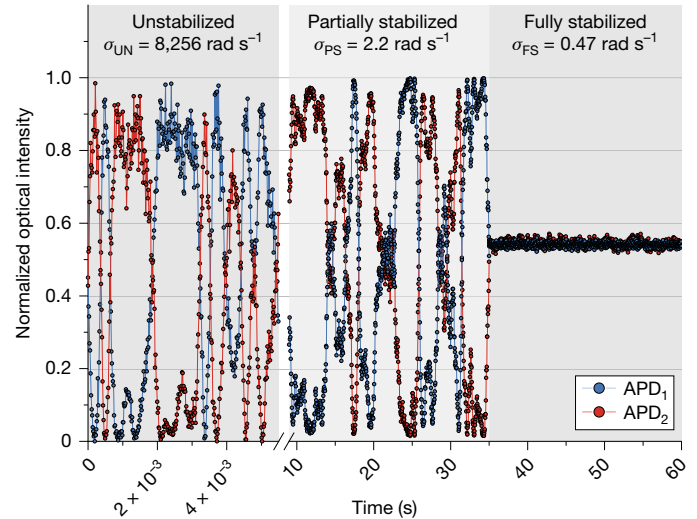


Fig. 4 | Phase stabilization of the channel. Optical interference between $\lambda_{\text{Q}}^{\text{A}}$ and $\lambda_{\text{Q}}^{\text{B}}$ measured by Charlie through APD₁ and APD₂ at three stages of the phase stabilization. During the unstabilized stage, the free-drifting interference between the signals is recorded. In the partially stabilized stage, data are captured when only coarse phase stabilization is activated. Full stabilization occurs when both coarse and fine phase stabilizations are activated. To accurately interpret the results, notice the interruption in the abscissa and the different timescales used. The increasing stability of the signal across the three stages can be characterized by the decrease in the average standard deviation of the phase drift.

experiment. This proves that our system effectively implemented an untrusted quantum repeater over commercial telecom infrastructure. Overcoming this bound, without using quantum memories⁵⁶ or photonic cluster states⁵⁷, and at parity of the detectors used, is only possible when optical coherence is maintained between the signals interfering at the central relay. Our result approached the relative SKC₀ (dashed line), and the simulations indicated that if all other parameters were conserved but Alice's and Bob's key generation links had been symmetric, the system would have overcome this bound. A detailed explanation of the plotted bounds and their analytical expression is provided in the Methods.

Figure 5 also compares our results with previous state-of-the-art QKD field trials using practical, non-cryogenic detectors. Our experiment demonstrates the benefits of leveraging optical coherence in extending the maximum distance of quantum communications, effectively doubling the distance achieved in previous trials and increasing the tolerable loss budget by approximately three orders of magnitude. The achieved SKR also compares favourably with the achieved rate sufficient to support low-rate one-time-pad encryption for critical data transfer or refresh Advanced Encryption Standard-256 (AES-256) keys every few seconds, more than enough to support operation of commercial-off-the-shelf AES encryptors.

By cooling the APDs to below -30°C , which is feasible using thermoelectric coolers⁵⁸, the system's range can be substantially extended. This improvement would enable it to surpass the state-of-the-art 71.9-dB link loss reported for cryogenically cooled point-to-point QKD systems⁵⁹. Further details are provided in the Supplementary Information.

The star-shaped topology of our system, with detectors situated at the central node, inherently facilitates network scalability by enabling straightforward expansion through the addition of transmitters connected to this central hub. A comprehensive analysis of the system's scalability for larger networks, alongside alternative approaches for phase stabilization, is provided in the Supplementary Information.

Our work demonstrates the compatibility of coherence-based quantum communications with existing network infrastructure,

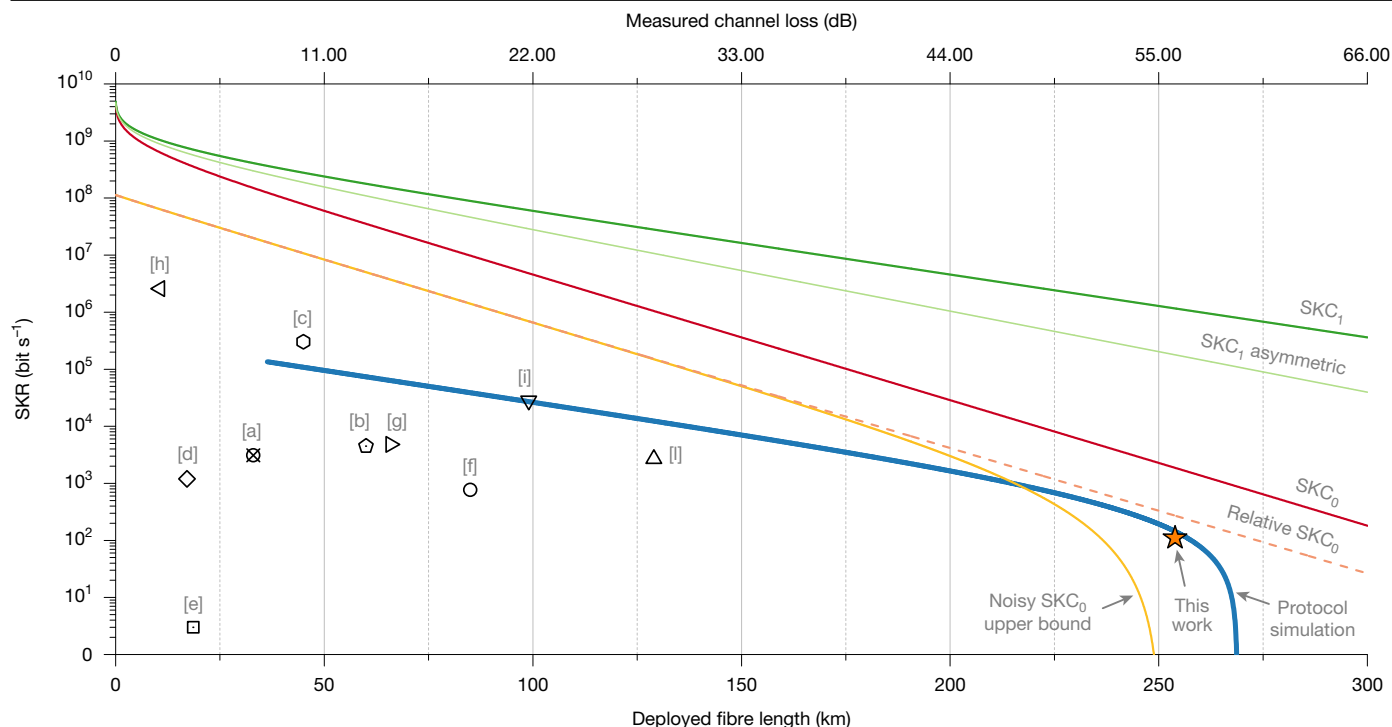


Fig. 5 | Secret key rate. The experimentally obtained SKR (star) and the simulation for our protocol (blue line) are plotted against the quantum channel length (bottom axis) and attenuation (top axis). The channel attenuation is calculated using the 0.22 dB km⁻¹ attenuation coefficient measured for this experiment. Other lines represent theoretical capacity bounds of quantum communication schemes, with analytical expressions provided in the Methods. SKC₁ is the bound for symmetric quantum communications aided by a single quantum repeater³⁴, and SKC₁ asymmetric represents the same bound for a channel where the repeater is positioned three-fifths of the way between the transmitters, as in our case. SKC₀ is the repeaterless secret key capacity bound³⁴,

with its relative version scaled down by the detectors' efficiency. The noisy SKC₀ upper bound represents the realistic upper bound capacity for repeaterless quantum communications using detectors with the same efficiency and dark counts as those used in this experiment. Open symbols represent the state-of-the-art SKR for field trial QKD tests using non-cryogenically cooled single-photon detectors plotted at the fibre distance reported by the experiments (with square-bracketed letters referencing the corresponding studies, listed in Supplementary Information section 1A). A more detailed discussion of these results, including a comparison with experiments conducted using SNSPDs, is provided in Supplementary Information section 1L.

and the practical implementation of an effective quantum repeater over commercial networks. We also achieved, to our knowledge, the longest distances for real-world QKD using non-cryogenically cooled detection technology and established a long-distance star-shaped QKD network with measurement-device-independent properties. Our findings confirm that the environmental conditions in operational telecommunication centres are comparable to or even better than those simulated in labs, encouraging further commercialization and prototyping of coherent quantum communication equipment. Utilizing practical solutions such as frequency dissemination and regeneration, off-band phase stabilization, and semiconductor-based equipment such as compact laser diodes and APDs, this demonstration sets a foundation for future implementations of advanced quantum communication protocols requiring coherence maintenance^{14,15}, such as Duan–Lukin–Cirac–Zoller-based quantum repeaters²⁸ and networks^{2–4}. This achievement, showcased over a highly asymmetric national-scale link, lays the groundwork for future high-performance, practical quantum communications and networking.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-08801-w>.

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Overview of the service and management layers

Simplified representations of the system and the signals' optical routing are reported in Figs. 1 and 2, respectively. A comprehensive schematic of the system is reported in the Supplementary Information.

The service layer, the top layer in the functional diagram in Fig. 1, comprises:

- An internet-connected virtual private network (VPN) server in Alice providing remote access to all the equipment connected to this layer.
- Two optical fibres, one per duplex, connecting Alice to Charlie (F_C^{AC}) and Bob to Charlie (F_C^{BC}). These are standard single-mode fibres, rented from the network provider, and injected with six optical signals at different wavelengths travelling in both directions (upstream and downstream). A full description of the signals transmitted over these fibres is reported in the Supplementary Information.
- Three optical MUX/DEMUX subsystems, one per node, injecting and extracting the optical signals into/from F_C^{AC} and F_C^{BC} .
- Three ethernet switches, one per node, endowed with RJ45 and small form-factor pluggable (SFP) transceiver ports to connect all the nodes' subsystems to the same local area network.
- A 125-MHz radio-frequency (RF) reference clock in Charlie, generated locally by FPGA^C and distributed optically to FPGA^A and FPGA^B, where the signal are converted back to RF.
- One seeding lasers subsystem in Charlie composed of two continuous-waveform lasers, L_S and L_Q , emitting at $\lambda_S = 1,551.72$ nm and $\lambda_Q = 1,550.12$ nm. Both lasers have an emission linewidth of a few hundred hertz. Outputs from L_S and L_Q are combined and sent to the transmitter nodes where they establish a common phase reference between Alice's and Bob's pulses through OIL.
- A 120-km-long fibre and an EDFA (shown in the detailed schematics in the Supplementary Information) to delay and amplify the λ_S and λ_Q signals for Bob before these are injected into F_C^{BC} . This delay ensures approximate matching of the overall optical path of Alice's and Bob's λ_S and λ_Q signals within the coherence length of their respective laser sources. This approach enables the execution of a coherence-based quantum communication protocol over a highly asymmetric channel and it could be removed using master laser sources with suitably narrow emission linewidth.

The management layer, the middle layer in the functional diagram shown in Fig. 1, consists of the following:

- Computer servers (Server^{A,B,C}): each node has a dedicated server connected to the same network by the service layer. These servers manage equipment at their respective nodes.
- FPGAs (FPGA^{A,B,C}): one FPGA per node ensures synchronization of the reference clock and absolute time among the nodes. The timing signals distributed by the service layer facilitate this synchronization. Notably, the jitter between the three node clocks of the nodes was tested in the lab and found to be less than 8 ps.
- Two AWGs: located at the transmitter nodes, these AWGs are locked to the reference clock provided by FPGA^A and FPGA^B. They generate gigahertz driving signals used by the quantum layer to encode the quantum communication protocol.

Overview of the quantum layer

The quantum layer, shown at the bottom of the functional diagram in Fig. 1 and in more detail in the Supplementary Information, comprises five subsystems: the quantum MUX/DEMUX unit and the quantum encoder at the transmitters, and the quantum receiver, four APDs and a time tagger at the receiver node.

Each quantum DEMUX/MUX unit implements the following operations:

- Amplification of the λ_S and λ_Q signals using an EDFA.

- Separation of λ_S from λ_Q using a frequency demultiplexer, and transmission of λ_Q to the quantum encoder.
- Adjustment of the intensity and polarization of the λ_S signal to stabilize the optical channel. The manipulated signal is labelled λ'_S .
- Multiplexing the tuned λ'_S signal with the encoded λ'_Q from the quantum encoder.
- Sending λ'_S and λ'_Q to Charlie by injecting them into the F_Q fibres connecting the transmitters to the receiver.

Each quantum encoder transforms the received λ_Q continuous-waveform signal into a λ'_Q signal, modulated according to the specifications of the SNS TF-QKD protocol. Within the encoder, λ_Q is injected into a laser diode tuned for optimal OIL with this signal. Detailed considerations regarding the implementation security aspects of using OIL in our system are provided in the Supplementary Information. The laser diode's output undergoes processing by 3 intensity modulators arranged in series, which carve the continuous-waveform signal into 1-GHz, 120-ps-long optical pulses modulated at 4 different intensities. Subsequently, two phase modulators placed in series modulate the phase of these pulses with arbitrary phase within the $[0, 2\pi)$ interval. These five high-speed electro-optics modulators are followed by a variable optical attenuator setting the average photon flux to the correct level, and an electronic polarization controller precompensating the polarization rotation introduced by the quantum channel.

When Alice's and Bob's modulated λ'_S and λ'_Q signals enter the quantum receiver, they each encounter a polarization beam splitter, which separates the light into horizontal (λ'^H) and vertical (λ'^V) polarization components. The four vertical polarization components $\lambda'^{A,V}$ and $\lambda'^{B,V}$ were connected by a 4-to-1 switch to APD₄ so that each could be monitored in turn to maximize the horizontal polarization components. The horizontal components $\lambda'^{B,H}$ from Bob pass then through a phase modulator, used to stabilize the optical interference. After this, $\lambda'^{A,H}$ and $\lambda'^{B,H}$ interfere on a polarization-maintaining beam splitter. The two outputs of the interference between $\lambda'^{A,H}$ and $\lambda'^{B,H}$ are monitored by APD₁ and APD₂. Clicks collected by these detectors are analysed by a time tagger, which provides the time-of-arrival information for each photon detection. The outputs of the interference between $\lambda'^{A,H}$ and $\lambda'^{B,H}$ can be monitored in turn by APD₃. The clicks recorded by this detector are used to implement the fast feedback of the off-band phase stabilization.

Off-band phase stabilization

In this technique, two optical signals close in frequency are multiplexed and transmitted together and thus pick up an almost identical phase noise introduced by the transmission channels. One of the two wavelengths, λ_S , acts as a support for the channel stabilization. λ_S is kept as a continuous-waveform signal and is modulated only in intensity and polarization to optimize the optical interference in Charlie between the λ'^A_S and λ'^B_S sent by Alice and Bob, respectively. By actively stabilizing at Charlie the interference between λ'^A_S and λ'^B_S through a fast, coarse, feedback loop driving a phase modulator, all the noise introduced by the channels is corrected at λ_S . The same phase correction is also applied to λ_Q , the wavelength used for the encoding of the quantum protocol, and this induces a reduction of more than three orders of magnitude to the phase noise picked up by λ_Q as well. The signal at λ_Q is fully stabilized by a second slower, fine, phase feedback acting on λ_Q alone and implemented through a fibre stretcher. This technique is instrumental to remove the phase noise introduced by the transmission channels, thus permitting the execution of a phase-based quantum communication protocol.

Actively odd-parity-pairing SNS TF-QKD protocol

In this experiment, we implemented the four-intensity SNS TF-QKD quantum communication protocol³⁰ over a highly asymmetric link⁵⁵. Our implementation incorporated the actively odd-parity

pairing (AOPP) technique^{47,60} during post-processing to enhance the signal-to-noise ratio. We also accounted for finite-size effects in our analysis.

The protocol involves repeating the following steps N_{tot} times, until sufficient raw key material is generated for post-processing.

Preparation of weak coherent states. Alice and Bob independently prepare a weak WCP of the form $|\sqrt{\mu} e^{i\theta}\rangle$, where the pulses intensity $|\mu|^2$ and phase θ are determined by the specific encoding chosen by the users.

Selection of code or test basis. Alice (Bob) first selects the code $\mathbb{Z}$, or test, or decoy $\mathbb{X}$ basis with probability $P_{\mathbb{Z}_A}$ ($P_{\mathbb{Z}_B}$) and $P_{\mathbb{X}_A}$ ($P_{\mathbb{X}_B}$), respectively, where $P_{\mathbb{Z}} + P_{\mathbb{X}} = 1$.

Pulse encoding. When the code basis $P_{\mathbb{Z}}$ is chosen, Alice (Bob) encodes a bit 1 (0) by preparing a phase-randomized WCP with intensity $|\mu|^2 = s_A$ ($|\mu|^2 = s_B$) with probability ϵ_A (ϵ_B), and encodes a bit 0 (1) by preparing a vacuum state $|\mu|^2 = n_A$ ($|\mu|^2 = n_B$) with probability $1 - \epsilon_A$ ($1 - \epsilon_B$). When the testing basis $P_{\mathbb{X}}$ is chosen, Alice (Bob) randomly selects a flux value $|\mu|^2 = \{u_A, v_A, w_A\}$ ($\{u_B, v_B, w_B\}$) with conditional probability $\{P_{u|\mathbb{X}_A}, P_{v|\mathbb{X}_A}, P_{w|\mathbb{X}_A}\}$ ($\{P_{u|\mathbb{X}_B}, P_{v|\mathbb{X}_B}, P_{w|\mathbb{X}_B}\}$), where $P_{u|\mathbb{X}} + P_{v|\mathbb{X}} + P_{w|\mathbb{X}} = 1$, and a random global phase value $\varphi \in [0, 2\pi)$. Owing to the asymmetry of the link, the encoding parameters that maximize the SKR in our experimental condition are different. In this scenario, the following relation between the encoding parameters ensure the protocol security as shown in ref. 55:

$$\frac{v_A}{v_B} = \frac{\epsilon_A(1 - \epsilon_B) s_A e^{-s_A}}{\epsilon_B(1 - \epsilon_A) s_B e^{-s_B}} \quad (1)$$

Pulse transmission and detection announcement. After the preparation stage, Alice and Bob send their encoded WCP to Charlie, who interferes them on a 50:50 beam splitter followed by two single-photon detectors monitoring its two outputs. When only a single detector clicks per interference, Charlie broadcasts which one clicked over a public communication channel. At this point, Alice and Bob check what basis they used for each successful detection announced by Charlie. They label events where they both encoded on the $\mathbb{X}$ basis as $\mathbb{X}$ windows, and events where they both encoded on the $\mathbb{Z}$ basis as $\mathbb{Z}$ windows.

Parameter estimation and SKR calculation. Events in the $\mathbb{X}$ windows are used for the security analysis. The phase error rate can be extracted from these events by post-selecting instances where Alice and Bob encoded the same decoy state (for example, u_A for Alice and u_B for Bob, or v_A and v_B), with the WCP satisfying the following phase matching condition:

$$|\theta_A - \theta_B| \leq \frac{2\pi}{M} \text{ or } |\theta_A - \theta_B - \pi| \leq \frac{2\pi}{M} \quad (2)$$

where θ_A and θ_B are the secret phases encoded by Alice and Bob, respectively, and M is the number of phase slices.

From their respective encoding associated with $\mathbb{Z}$ windows, Alice and Bob each extract a raw key of length n_r (the total raw key). The discrepancies between these two strings constitute the bit-flip error rate E_Z . The AOPP method is a post-processing technique requiring two-way classical communication between Alice and Bob. It reduces E_Z at the cost of reducing the length of n_r . In AOPP, Bob pairs up all the bits 0 in his raw strings with bits 1. He then communicates to Alice the positions of the pairs. Alice then pairs the bits in the same manner as Bob and checks their parity. She then discards all the resulting pairs with even parity, and communicates the position of the surviving pairs to Bob. They then discard the second bit of the surviving pairs from their strings and each creates a second, shorter, string n'_r with a considerably smaller

bit-flip error rate E'_Z , which they will use to extract the final key. This process enables longer-distance communication with higher SKR.

The decoy-state method^{51,52} enables precise estimation of the number of single-photon quantities needed for the SKR calculation through analysis of the events recorded in the $\mathbb{X}$ windows. In the following, we denote the counting rate associated with the pulses of intensity μ_A ($\mu_A = u_A, v_A, w_A$) for Alice and μ_B ($\mu_B = u_B, v_B, w_B$) for Bob as $S_{\mu_A, \mu_B} = n_{\mu_A, \mu_B} / N_{\mu_A, \mu_B}$, where N_{μ_A, μ_B} is the number of pulse pairs of source μ_A, μ_B sent out in the whole protocol, and n_{μ_A, μ_B} is the total number of one-detector heralded events of source μ_A, μ_B . S_{μ_A, μ_B} can be measured experimentally. Using the recorded count rate, and the information on the photon fluxes, it is possible to estimate the lower bounds of the expected values of the counting rate of states $|01\rangle$ $|01\rangle$ and $|10\rangle$ $|10\rangle$, which are

$$\langle s_{01} \rangle = \frac{u_B^2 e^{v_B} \langle s_{w_A, v_B} \rangle - v_B^2 e^{u_B} \langle s_{w_A, u_B} \rangle - (u_B^2 - v_B^2) \langle s_{w_A, w_B} \rangle}{u_B v_B (u_B - v_B)} \quad (3)$$

$$\langle s_{10} \rangle = \frac{u_A^2 e^{v_A} \langle s_{v_A, w_B} \rangle - v_A^2 e^{u_A} \langle s_{u_A, w_B} \rangle - (u_A^2 - v_A^2) \langle s_{w_A, w_B} \rangle}{u_A v_A (u_A - v_A)} \quad (4)$$

where $\langle \ast \rangle$ and $\langle \bar{\ast} \rangle$ represent the lower and the upper bound of the corresponding expected values, with a composable definition of security and the Chernoff bound being applied. The lower bound of the expected value of the counting rate of untagged (that is, single) photons is then:

$$\langle s_1 \rangle = \frac{v_A}{v_A + v_B} \langle s_{10} \rangle + \frac{v_B}{v_A + v_B} \langle s_{01} \rangle \quad (5)$$

The lower bounds of the expected value of the number of untagged 0 bits, $\langle n_{01} \rangle$, and untagged 1 bits, $\langle n_{10} \rangle$, can be calculated as:

$$\langle n_{01} \rangle = N_{\text{tot}} P_{\mathbb{Z}_A} P_{\mathbb{Z}_B} \epsilon_A (1 - \epsilon_B) s_A e^{-s_A} \langle s_{01} \rangle \quad (6)$$

$$\langle n_{10} \rangle = N_{\text{tot}} P_{\mathbb{Z}_A} P_{\mathbb{Z}_B} \epsilon_B (1 - \epsilon_A) s_B e^{-s_B} \langle s_{01} \rangle \quad (7)$$

The expected value of the lower bounds of the number of untagged bits is:

$$\langle n_1 \rangle = \langle n_{01} \rangle + \langle n_{10} \rangle \quad (8)$$

The phase-flip error rate of untagged bits in $\mathbb{Z}$ windows, $\langle e_1^{\text{ph}} \rangle$, can be calculated from the bit-flip error rate of untagged bits in $\mathbb{X}$ windows as

$$\langle e_1^{\text{ph}} \rangle = \frac{\langle \bar{T}_{X_1} \rangle - e^{-v_A - v_B} \langle s_{w_A, w_B} \rangle / 2}{e^{-v_A - v_B} (v_A + v_B) \langle s_1 \rangle} \quad (9)$$

where $\langle \bar{T}_{X_1} \rangle$ is the ratio between the total number of pulses sent out with intensities v_A and v_B , and the number of corresponding error events.

The SKR of finite-size SNS TF-QKD with AOPP is calculated as:

$$R = \frac{2}{N_{\text{tot}}} \{n'_1 [1 - h(e_1^{\text{ph}})] - f_{\text{EC}} n'_r h(E'_Z) - \Delta_{\text{FS}}\} \quad (10)$$

where $h(x) = -x \log_2 x - (1-x) \log_2 (1-x)$ is the Shannon entropy, $f_{\text{EC}} = 1.05$ the error correction efficiency factor, and $\Delta_{\text{FS}} = \log_2 \frac{2}{2\epsilon_{\text{cor}}} + 2 \log_2 \frac{1}{\sqrt{2\epsilon_{\text{PA}} \hat{\epsilon}}}$ is the finite-size correction term, with $\epsilon_{\text{cor}} = 10^{-10}$ the error correction failure probability, $\epsilon_{\text{PA}} = 10^{-10}$ the privacy amplification failure probability, and $\hat{\epsilon} = 10^{-10}$ the smoothing parameter for the min- and max-entropy chain rules. n'_1 is the number of the untagged bits after AOPP, and e_1^{ph} is their phase-flip error rate.

Further details regarding the SKR calculation process can be found in ref. 47, which was closely followed in this study's computations, mirroring methodologies used in previous experimental works^{36,38,42,61}.

Secret key capacity bounds

Our quantum communications system consists of a simple quantum network composed of two end users, Alice (A) and Bob (B), and a middle user, Charlie (C). The network spans between the distance between Frankfurt and Kehl, with the central relay positioned in Kirchfeld at an asymmetric distance, approximately three-fifths of the total span.

The two-way assisted capacity $\mathcal{C}$ determines the optimal end-to-end rate of quantum communications achievable without restrictions on local operations and classical communication³⁴. For QKD, the SKC_0 is equal to $\mathcal{C}$ (ref. 34). Therefore, the end-to-end capacity and secret key capacity of a simple asymmetric one-repeater network for pure-loss channels is given by⁶²

$$\text{SKC}_1(\eta_A, \eta_B) = -\log_2[1 - \min(\eta_A, \eta_B)] \text{ ebits per use}, \quad (11)$$

where η_A and η_B are the transmissivities of the lossy channels separating Alice and Bob, respectively, from Charlie. The subscript in the notation SKC_1 refers to the number of quantum repeaters. The channel transmissivities can be asymmetric, that is, when $\eta_A \neq \eta_B$. These capacities are plotted in Fig. 5. We plot SKC_1 for a centred Charlie (green) and SKC_1 asymmetric for an asymmetric configuration as in our system (light green).

At low transmissivities, the single-repeater capacity scales like the worst transmissivity of the network, that is, $\text{SKC}_1(\eta_A, \eta_B) \approx 1.44 \min(\eta_A, \eta_B)$. Although the highest rate of any implementation of TF-QKD cannot overcome the single-repeater capacity, TF-QKD can achieve the optimal rate–distance scaling, up to a constant factor. This optimal scaling can be observed in Fig. 5 for the simulation of our TF-QKD implementation (blue).

By contrast, the rate of any point-to-point quantum communication protocol scales like the total transmissivity with distance⁶³. The fundamental rate–distance limit without a repeater is given by the following³⁴

$$\text{SKC}_0(\eta_{\text{channel}}) = -\log_2(1 - \eta_{\text{channel}}) \text{ ebits per use}, \quad (12)$$

where $\eta_{\text{channel}} = \eta_A \eta_B$ is the total channel transmissivity of the lossy channel separating Alice and Bob. This two-way assisted capacity represents the ultimate rate that is reachable without a quantum repeater and is plotted in Fig. 5 (red). The rate of our system (star marker) closely approaches the $\text{SKC}_0(\eta_{\text{channel}})$ without overcoming it. However, to assess the advantages of adopting a coherence-based QKD protocol over other point-to-point repeaterless QKD implementations, we benchmark our results against the upper bound of the maximum rate achievable by a point-to-point system using detectors with similar properties (efficiency and dark counts) to those used in this experiment.

The upper bound on the noisy repeaterless capacity for a channel transmissivity $\eta_{\text{channel}} = \eta_A \eta_B$, detector efficiency ϵ , and dark-count rate d_c , is given by the upper bound on the thermal-loss channel capacity $\text{SKC}_0^{\text{noisy}}(\eta_{\text{total}}, \bar{n})$ characterized by total transmissivity $\eta_{\text{total}} = \eta_{\text{channel}} \epsilon$ and mean photon number $\bar{n} \approx d_c / (1 - \eta_{\text{total}})$, that is³⁴

$$\text{SKC}_0^{\text{noisy}}(\eta_{\text{total}}, \bar{n}) \leq \text{UB}(\eta_{\text{total}}, \bar{n}) \text{ ebits per use}, \quad (13)$$

where $\text{UB}(\eta_{\text{total}}, \bar{n}) = -\log_2[(1 - \eta_{\text{total}})\eta^{\bar{n}}] - h(\bar{n})$ for $\bar{n} < \eta_{\text{total}} / (1 - \eta_{\text{total}})$, and $h(x) \equiv (x+1)\log_2(x+1) - x\log_2(x)$. We plot this noisy SKC_0 upper bound in Fig. 5 (solid yellow). A repeater is required to overcome this upper bound assuming Alice and Bob use non-ideal detectors with untrusted efficiency ϵ and dark-count rate d_c . Figure 5 clearly shows that our system produces an SKR higher than the noisy SKC_0 upper bound, meaning that no point-to-point protocol using similar detectors can achieve our rate. We derive this result in the following paragraphs.

We first model detectors with non-unit efficiency ϵ by including additional loss before the detector with transmissivity ϵ (ref. 64), then the capacity adjusted for inefficient detectors becomes

$$\text{SKC}_0(\eta_{\text{total}}) = -\log_2(1 - \eta_{\text{total}}) \text{ ebits per use}, \quad (14)$$

where $\eta_{\text{total}} = \eta_{\text{channel}} \epsilon$ (pink dashed line in Fig. 5).

Still, the capacity assumes no additional sources of noise. The primary source of noise other than pure loss comes from detector dark counts that are false detection events at the detectors. We can simultaneously model dark counts and non-ideal efficiency by including a Gaussian thermal-loss channel before each detector⁶⁴. That is, we assume that the dark counts are originating in the channel, and for any channel transmissivity η_{channel} , detector efficiency ϵ and dark-count rate d_c , there is an equivalent thermal-loss channel characterized by the total transmissivity η_{total} and mean photon number $\bar{n}$. Although the exact achievable capacity of the thermal-loss channel is unknown, an upper bound is known³⁴ and is given in equation (13).

The thermal-loss channel with transmissivity η_{total} and mean photon number $\bar{n}$ is equivalent to interfering a thermal state with mean photon number $\bar{n}$ with the input state on a beam splitter of transmissivity η_{total} , where the thermal state is defined as

$$\rho_{\bar{n}} = \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(\bar{n}+1)^{n+1}} |n\rangle \langle n|. \quad (15)$$

Then, the dark-count rate is defined as the probability the detector registers 'ON' when no photons are input onto it, that is, when the input state is vacuum.

As our detectors have low dark-count rates ($d_c < 10^{-6}$), this translates to a small value of $\bar{n}$. To first order in $\bar{n}$, the thermal state is

$$\rho_{\bar{n}} \approx \frac{1}{\bar{n}+1} |0\rangle \langle 0| + \frac{\bar{n}}{(\bar{n}+1)^2} |1\rangle \langle 1| \approx |0\rangle \langle 0| + \bar{n} |1\rangle \langle 1|. \quad (16)$$

That is, the photon is prepared with probability $\frac{\bar{n}}{(\bar{n}+1)^2} \approx \bar{n}$ and reflected off the beam splitter with probability $1 - \eta_{\text{total}}$ giving the probability to be detected 'ON' at the detector

$$P(\text{'ON'}) = d_c \approx (1 - \eta_{\text{total}}) \bar{n}. \quad (17)$$

Thus, to first order in $\bar{n}$ we find that

$$\bar{n} \approx \frac{d_c}{1 - \eta_{\text{total}}}. \quad (18)$$

This approximation is sufficient for presenting results. Numerically computing $\bar{n}$ to higher order in photon number has no visually noticeable change to the upper bound plotted in Fig. 5.

Data availability

All data supporting the findings of this study are available in the paper and its Supplementary Information. Additional details can be obtained from the corresponding author upon request.

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Additional information

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